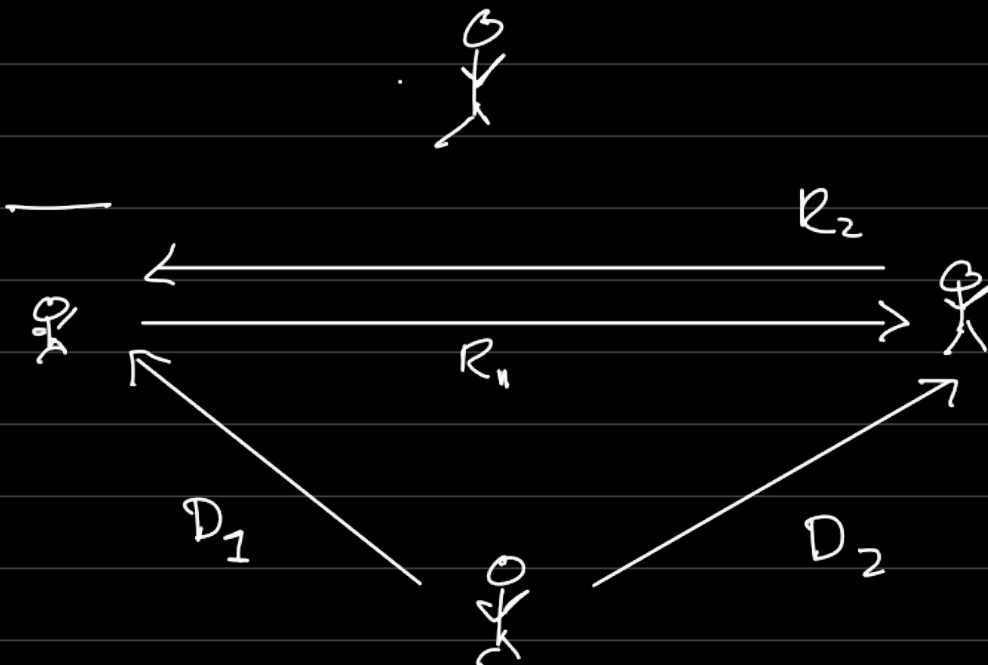
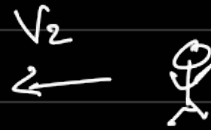


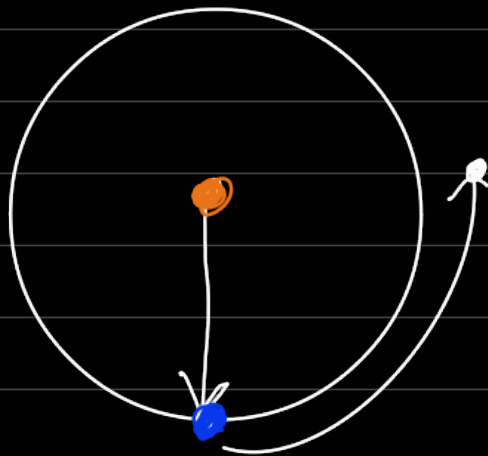
* Reference frames:

- inertial: non-accelerating
- non-inertial: accelerating

*



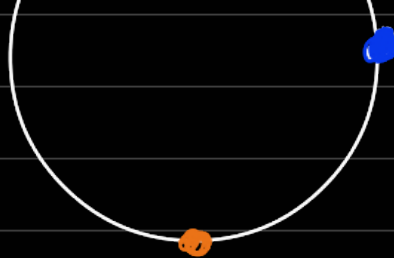
$$* \vec{r} = (x(t), y(t)) = (0, -Ut)$$



$$\vec{v} = (0, -U)$$

$$\vec{a} = (0, 0)$$





$$\frac{d\varphi}{dt} = \omega(t)$$

$$* \quad \frac{d\vec{r}}{dt} = \vec{\omega} \times \vec{r}$$

ω : constant.

$$\bullet \quad \frac{d\hat{e}_\xi}{dt} = \vec{\omega} \times \hat{e}_\xi$$

$$\vec{r}_s = x(t)\hat{e}_x + y(t)\hat{e}_y + z(t)\hat{e}_z$$

$$\vec{r}_\Sigma = \xi(t)\hat{e}_\xi(t) + \eta(t)\hat{e}_\eta(t) + \zeta(t)\hat{e}_\zeta(t)$$

$$\vec{v}_s = \dot{x}(t)\hat{e}_x + \dots$$

$$= v_{\Sigma,app} + \vec{\omega} \times \vec{r}_\Sigma$$

$$v_{\Sigma,app} = \dot{\xi}\hat{e}_\xi + \dot{\eta}\hat{e}_\eta + \dot{\zeta}\hat{e}_\zeta$$

$$a_{\Sigma,app} = \ddot{\xi}\hat{e}_\xi + \ddot{\eta}\hat{e}_\eta + \ddot{\zeta}\hat{e}_\zeta$$

$$* \quad \vec{\omega} = \omega\hat{e}_z$$

$$\vec{\omega} \times \vec{r}_{\Sigma,app} = \omega\dot{\xi}(-\hat{e}_\eta) + \dot{\eta}\omega\hat{e}_\xi$$

$$\vec{\omega} \times \vec{r} = \dot{\xi}(t)\omega(-\hat{e}_\eta) + \dot{\eta}\omega\hat{e}_\xi$$

$$\begin{aligned} \vec{\omega} \times (\vec{\omega} \times \vec{r}) &= \dot{\xi}(t)\omega^2\hat{e}_\xi + \dot{\eta}\omega^2\hat{e}_\eta \\ &= \omega^2(\dot{\xi}(t)\hat{e}_\xi(t) + \dot{\eta}(t)\hat{e}_\eta(t)) \end{aligned}$$

• Cent. force + Coriolis

$$* \quad \ddot{\xi} = \omega^2 \xi(t) + 2\omega\dot{\eta}$$

$$i\ddot{\eta} = i(\omega^2 \eta - 2\omega \dot{\xi})$$

$$\ddot{z} = \ddot{\xi} + i\ddot{\eta}$$

$$\ddot{\xi} + i\ddot{\eta} = \omega^2(\xi + i\eta) + 2\omega(\dot{\eta} - i\dot{\xi})$$

$$\ddot{z} = \omega^2 z - 2\omega i \dot{z}$$

$$\ddot{z} + 2\omega i \dot{z} - \omega^2 z = 0$$

$$z(t) = e^{i\alpha t}$$

$$\dot{z} = i\alpha z$$

$$\ddot{z} = -\alpha^2 z$$

$$(-\alpha^2 + 2\omega i(i\alpha) - \omega^2) = 0$$

$$\alpha^2 + 2\omega\alpha + \omega^2 = 0$$

$$\alpha = \frac{-2\omega \pm \sqrt{4\omega^2 - 4\omega^2}}{2} = \alpha = -\omega$$

$$= A \cos(\omega t) - iA \sin(\omega t)$$

$$z(t) = A(t) e^{-i\omega t}$$

$$\dot{z} = \dot{A} e^{-i\omega t} - A i\omega e^{-i\omega t}$$

$$\ddot{z} = \ddot{A} e^{-i\omega t} - \dot{A} i\omega e^{-i\omega t} - \dot{A} i\omega e^{-i\omega t} + A(i\omega)^2 e^{-i\omega t}$$

$$(\ddot{A} - 2i\omega\dot{A} - \omega^2 A) + 2i\omega(\dot{A} - A i\omega) - \omega^2 A = 0$$

$$\ddot{A} - 2i\omega\dot{A} - \omega^2 A + 2i\omega\dot{A} + 2\omega^2 A - \omega^2 A = 0$$

$$\ddot{A} = 0$$

$$A(t) = Bt + C$$

$$A(0) = 0$$

$$C = 0$$

$$A(t) = Bt$$

$$z(t) = Bt e^{-i\omega t} = B e^{-i\omega t} (t + i t^2)$$

$$\xi(t) = Bt \cos(\omega t)$$

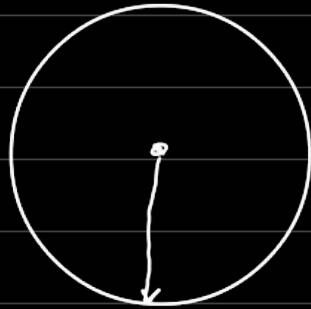
$$\eta(t) = -Bt \sin(\omega t)$$

$$|r|^2 = U^2 t^2 = t^2 B^2$$

$$B = \pm U$$

$$\xi(t) = Ut \cos(\omega t)$$

ω :



$$\theta = \frac{\pi}{4}$$

$2\pi R = \text{Distance}$

T : time

Rot. / sec

$$d\theta/dt = \omega$$

$$U = \frac{D}{T}$$

$$\omega T$$

$$\theta = \omega$$

$$U = \frac{R}{T/4} = \frac{\pi \omega}{2 \cdot 2\pi R} = \frac{\omega}{4R} \quad \frac{\pi}{2\omega} = \frac{\omega T}{4}$$

$$= \frac{2\omega R}{\pi}$$

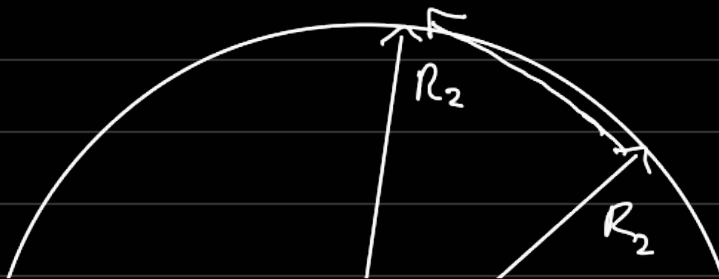
$$\dot{\xi}(t) = B \cos(\omega t) - Bt \sin(\omega t)$$

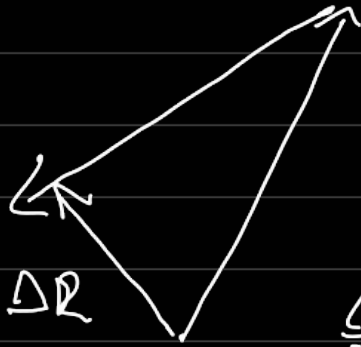
$$\dot{\xi}(0) = B = -U$$

$$\dot{\eta}(0) = -B \sin(\omega t) - Bt \cos(\omega t) = 0$$

$$\xi(t) = -U \cos(\omega t)$$

$$+ \omega \times r = \omega$$

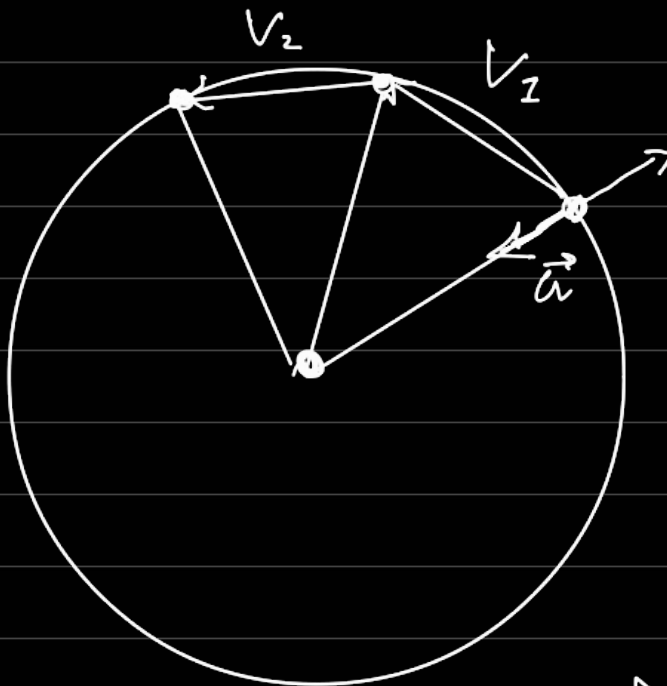




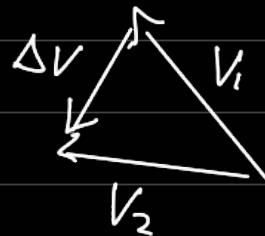
$$R_1 + \Delta R = R_2$$

$$\frac{\Delta R}{\Delta t} = \frac{\vec{R}_2 - \vec{R}_1}{\Delta t} = \vec{v}$$

a



$$\frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{\Delta t} = \vec{a}$$



$$F = -ma = T$$

$$v = \frac{2\pi R}{T_p}$$

$$v = \omega r$$

$$\omega T_p = 2\pi$$

$$\omega = (2\pi / T_p)$$

$$s = r\theta$$

$$\frac{ds}{dt} = r \frac{d\theta}{dt} = r\omega = v$$

